CLEMENT CHUNG

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Education

University of Waterloo **Waterloo, ON**
Bachelor of Computer Science, Sep 2025 – May 2030
- cGPA: 4.0 + B.P. Dammizio Entrance Scholarship

Skills

Programming: C++, CUDA, Rust, Python, Java, TypeScript
Software Tools: Linux, CMake, Docker, Git, CUDA, WebRTC

Experience

Systems Software Intern **May 2026 – Sep 2026**
Martinrea International *Vaughan, ON*
- Built C++ fuzz tests that generated malformed configuration and project files, automatically capturing parser crashes so they could be reproduced in CI.
- Integrated **Crashpad** into the C++ build and runtime, including a custom vcpkg port, Clang compatibility patch, packaged crash handler, and process-level crash reporting.
- Stabilized a flaky Linux CI integration test by correcting process exit-code propagation, signal handling, and data races, achieving a **100% observed pass rate**.

Real-Time Voice AI Intern **Jul 2025 – Aug 2025**
AI Stealth Startup *Mountain View, CA*
- Architected a dual-agent voice pipeline that ran retrieval and API workloads independently from token generation, reducing end-to-end response latency **6x, from 1.5 s to 250 ms**.
- Removed neural phoneme inference from the real-time media path and processed **10 ms WebRTC audio frames** with deterministic signal analysis, achieving near-zero lip-sync latency.

Projects

Qwen3-4B Quantized Inference Engine | C++, CUDA
- Implemented a quantized Qwen3-4B inference engine with GGUF model loading, KV caching, and custom Q4_K/Q6_K CUDA dequantization kernels.
- Benchmarked custom CUDA kernels at **63 tokens/s** - approximately **91% of llama.cpp throughput** - with automated correctness tests.

MiniTokenizer | Rust, Python, BPE
- Implemented BPE training and encoding in Rust, updating only token pairs affected by each merge instead of rescanning the full sequence.
- Added a Python API, UTF-8 support, and JSON serialization; measured encoding at **1.68 ms / 19.96 MB/s** and decoding at **0.75 ms / 44.48 MB/s**.

Mathphic | Next.js, FastAPI
- Built a full-stack application that transforms math questions into narrated, interactive visualizations using a Next.js frontend and FastAPI backend.
- Won **Most Technically Complex AI Hack** at Hack Canada 2026 among **140 participants**.

Awards

- **Hack the 6ix 2026 - QNX Best Use of QNX + Top 8 Finalist:** Integrated camera tracking with UART motor control for an autonomous turret running on QNX and Raspberry Pi.

Activities

Waterloo Virtual Reality Labs - Firmware Team Member | Oct 2025 - Present
- Developed Linux display-driver and Monado/OpenXR integration for a Raspberry Pi 5-based VR operating system.